IMU, 1750 IMU, 1775 IMU (Fig. 17), CG-5100 IMU and TG-6000 IMU, offer bias stability specification of better than 0.05°/hr (for 1775 IMU) and maximum input rate specification of ±490° (for 1775 IMU).

Military applications of laser gyroscopes

The technology of laser gyroscopes including both RLGs and FOGs has come of age. Laser gyroscopes are more than a laboratory experiment. Today, these are very strong candidates for being the preferred INS for almost all military and commercial applications, including military and commercial aircraft, tactical and strategic missiles, naval and marine vehicles, land vehicles and weapon platforms, and spacecraft.

Different application areas demand different levels of gyroscopic accuracy. Acceptable range of angular rate error for various military-grade applications employing laser gyroscopes for navigation and guidance is 0.05°/hr - 1.0°/hr (short-range tactical missiles), 0.005°/hr - 0.1°/hr (strategic cruise missiles), 0.0001°/hr - 0.01°/hr (aircraft inertial navigation) and 0.00005°/hr - 0.005°/hr (ballistic missile guidance).

Some representative examples of commercial aircraft using RLG INS include Airbus-320, Airbus-330/340, Airbus-380, Boeing 757-200 and Boeing-777.

RLG INS is also used in fighter aircraft such as F-15E Strike Eagle, F-16 Fighting Falcon, F/A-18 combat jets, F-22 Raptor, Sukhoi-Su-30MKI, HAL-Tejas and B-52H with AMI upgrade, MC-130E Combat Talon-I and MC-130H Combat Talon-II special-mission aircraft, EF-111 Raven electronic warfare aircraft, MH-60R, MH-60S, SH-60F and SH-60B Seahawk helicopters, Trident-I and Trident-II submarine-launched ballistic missiles, Shaurya hypersonic surface-to-surface tactical missile, AGNI-III and AGNI-IV intermediate range ballistic missiles, AGNI-V and AGNI-VI intercontinental ballistic missiles and International Space Station (ISS).

A320, A330, A340 and A380 families of passenger aircraft from Airbus Industries and Boeing's 737, 757 and 777 family passenger aircraft use Honeywell's ADIRS. Military aircraft including F-15E, F-16 and F/A-18 also use Honeywell's GG1320 digital RLG-based INS.

Sukhoi Su-30MKI fighter aircraft employs SAGEM's Sigma-95 integrated GPS and digital laser gyroscope INS.

Northrop Grumman's LTN-101 Flagship is also in use in a large number of commercial aircraft of single-aisle and long-range families from Airbus Industries.

SAAB-2000 twin-engine turboprop passenger aircraft built by SAAB, Sweden also uses LTN-101 navigation system.

Northrop Grumman's LN-100G INS is also used on a wide range of military platforms, including fighter aircraft (F-22 Raptor of Lockheed Martin), trainer aircraft (T-45 Goshawk of Boeing and BAE Systems), transport aircraft (C-130 of Lockheed Martin), pods, missiles, UAVs and unmanned under sea vehicles and helicopters (MH-60 series Seahawk helicopters).

AGNI-III and AGNI-IV intermediate range ballistic missiles and AGNI-V intercontinental ballistic missile (ICBM) in service since 2014 use indigenously-developed RLG-based INS optionally augmented by GPS terminal guidance with possible radar scene correlation.

AGNI-VI ICBM, likely to be inducted in service from 2018-19, will also use RLG-based INS augmented by GLONASS/IRNSS with possible radar scene correlation.

Canister-launched hypersonic surface-to-surface tactical missile uses indigenously-developed RLG-based INS.

Trident-II (UGM-133 D5) submarine-launched intercontinental ballistic missile in use by the UK on Vanguard and the US on Ohio class submarines also employs RLG-based INS. It, along with a stellar reference system, provides CEP of 90 m.

FOG-based 1775 IMU is used in a wide range of applications such as turret and weapon platform stabilisation and pointing, antenna platform positioning and stabilisation, mobile mapping systems, and stabilisation systems for radar, LIDAR and high-speed gimbals.

To be continued